

Resonance Quantum Theory (RQT): A Framework for Resonance-Base Structure and Interaction Conceptual Foundations and the Roton Quantum Model

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Abstract

The Resonance Quantum Theory (RQT) is a conceptual framework that interprets physical structure, interaction, and persistence as consequences of resonance organization rather than as properties imposed through fundamental particles, fields, or quantization rules. The framework begins from the assumption that localized resonance relationships can form within a continuous resonance-supporting continuum, giving rise to stable structures, interaction channels, and hierarchical organization across multiple scales.

In RQT, observable physical entities are viewed as dynamically maintained resonance structures. Stability arises through resonance closure, while interactions emerge through open and residual resonance channels connecting otherwise localized systems. Information, inertia, and binding are interpreted as consequences of the accommodation and reconfiguration of these resonance relationships. Matter and radiation are therefore treated not as fundamentally distinct categories, but as different manifestations of resonance organization within a common underlying framework.

The theory further proposes that physical structure develops hierarchically. Larger and more persistent systems emerge from networks of mutually stabilizing resonance configurations, while observable forces reflect residual effects of incomplete resonance closure. In this view, quantization, stability, and long-range interaction arise as natural consequences of resonance geometry and organization rather than as independent postulates.

To facilitate visualization, analysis, and numerical investigation, the Roton Quantum Model (RQM) is introduced as a discrete structural representation of the broader RQT framework. RQM projects resonance structures onto simplified resonance centers and channels, allowing complex resonance relationships to be described through a tractable geometric model while preserving the central concepts of closure, hierarchy, and interaction.

The purpose of this paper is to establish the conceptual foundations of RQT, define its principal structural concepts, introduce the relationship between RQT and RQM, and explore how resonance-based organization may provide a unified description of physical structure across scales.

1 Introduction

Nature exhibits persistent structure across an extraordinary range of scales. Atomic systems form stable molecules, molecules organize into complex materials, and larger structures emerge through increasingly intricate patterns of interaction. Despite their diversity, such systems share a common characteristic: they maintain coherence, stability, and recognizable identity while continuously interacting with their surroundings.

Modern physics provides highly successful mathematical descriptions of these phenomena. Quantum theory, quantum field theory, and relativity accurately predict a vast range of experimental observations and have established a remarkably consistent framework for understanding physical behavior. At the same time, many fundamental properties of physical systems are introduced through independent postulates, parameters, symmetries, or interaction rules. Stability,

quantization, binding, charge, and long-range interaction are typically described within specialized theoretical frameworks, each optimized for a particular domain of observation.

This raises a broader conceptual question: might these apparently distinct phenomena arise from a common underlying organizational principle? Rather than beginning with particles, fields, or forces as fundamental entities, one may instead ask whether physical structure itself emerges from the formation and persistence of relationships between interacting systems.

The Resonance Quantum Theory (RQT) explores this possibility. The framework is based on the assumption that resonance relationships can form between localized structures and that these relationships may themselves become stable, persistent, and capable of further interaction. In this view, physical systems are understood not primarily as isolated objects but as dynamically maintained networks of resonance relationships. Stability, interaction, and hierarchy are interpreted as consequences of the organization of these relationships across different scales.

The purpose of RQT is not to replace successful physical descriptions, but to investigate whether a common resonance-based framework can provide a deeper conceptual basis from which a wide range of observed phenomena may emerge. The theory therefore focuses on the formation of resonance structures, the development of stable interaction channels, and the mechanisms through which increasingly complex systems can arise from comparatively simple assumptions.

To facilitate visualization and quantitative exploration of these concepts, a simplified structural representation referred to as the Roton Quantum Model (RQM) is introduced later in this work. RQM provides a discrete geometric description of resonance structures and their interactions while remaining rooted in the broader conceptual framework established by RQT.

The following chapters develop the fundamental concepts of the theory, introduce the notion of resonance channels and closure, and examine how physical structure may emerge from the organization of resonance relationships across multiple levels of complexity.

2 Conceptual Foundations

The conceptual foundation of RQT is intentionally minimal. Rather than beginning with particles, fields, or forces, it starts from the possibility that stable identities may emerge through persistent relationships. Once such an identity has formed, it may itself participate in further relationships, allowing increasingly complex structures to arise through repeated application of the same principle. Figure 1 summarizes this recursive concept, which is developed throughout the remainder of this section.

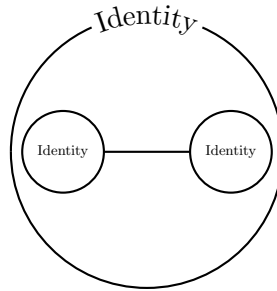


Figure 1: Recursive Formation of Identity

The remainder of this section develops the individual conceptual elements of this recursive process.

2.1 Persistence and Self-Resonance

Before a structure can interact, it must first persist.

The Resonance Quantum Theory begins with a minimal conceptual assumption: the existence of a continuum capable of supporting oscillatory behavior. The framework does not initially specify the origin, composition, or geometry of this continuum. It merely assumes that localized oscillatory configurations may arise within it.

An oscillatory configuration does not necessarily constitute a persistent structure. Of particular interest are configurations that repeatedly return to compatible states. Such recurrence allows an oscillation to maintain its identity over extended durations and establishes the possibility of persistence.

Within RQT, persistence is regarded as the first prerequisite for the emergence of structure. A persistent oscillatory configuration forms a self-maintaining cycle whose future evolution remains compatible with its preceding states. This process is referred to as self-resonance. At the conceptual level adopted here, self-resonance may be regarded as the simplest form of persistent resonance behavior.

The precise nature of such self-resonant configurations is not specified at this stage. Their significance lies not in their detailed structure, but in the fact that they provide a mechanism through which stable and persistent patterns may arise. Once persistence is established, increasingly complex forms of coupling, interaction, and organization become possible.

RQT therefore considers persistent self-resonant configurations as the conceptual starting point from which higher-order resonance structures can emerge.

2.2 Stability and Identity

Persistence allows oscillatory behavior to remain. Stability allows it to become structure.

A persistent oscillatory configuration does not necessarily remain unchanged. Local adjustments, interactions, and ongoing accommodation may continuously influence its evolution. Within RQT, stability therefore does not imply rigidity or immutability. Instead, a structure is considered stable when it repeatedly returns to sufficiently compatible states, allowing its characteristic behavior to persist despite ongoing change.

Such stable resonance structures represent the first distinguishable configurations within the framework. Their identity does not arise from a fixed internal composition, but from the persistence of recognizable resonance properties over time. A stable structure may therefore be characterized through its phase relationships, recurrence behavior, and interaction potential rather than through a permanently defined geometry.

The continued existence of a stable resonance structure also preserves information about its formation and evolution. This information is not regarded as a separately stored quantity, but as an intrinsic consequence of the ongoing resonance process itself. The present state of a structure remains connected to preceding states through a continuous resonance flow, allowing distinctions and relationships to persist over time.

Once stable resonance structures emerge, they may participate in increasingly complex forms of coupling and organization. Stability therefore provides the foundation upon which interaction, relationship formation, and higher-order resonance structures can develop.

Within RQT, stable resonance structures constitute the first identifiable building blocks from which larger resonance networks and hierarchical organizations may arise.

2.3 Interaction and Relationship Formation

Interaction creates relationships. Relationships create shared identity.

Stable resonance structures do not exist in complete isolation. Whenever the state of one structure influences the evolution of another, an interaction is established. The precise mechanism through which this influence occurs is not specified at this stage. It is sufficient to recognize that persistent structures may affect one another and thereby participate in a common process of development.

Within RQT, interactions are significant not merely because they modify existing structures, but because they give rise to relationships. A relationship represents an organized connection between two or more structures whose future evolution can no longer be described entirely independently. Through interaction, participating structures begin to share aspects of their resonance behavior and thereby become part of a larger organizational whole.

The emergence of a relationship introduces an additional level of identity. While the participating structures may remain individually distinguishable, the relationship itself acquires significance as a persistent element of the overall resonance organization. From a higher-level perspective, the relationship may therefore be regarded as a structure in its own right.

Relationships need not possess equal strength or persistence. Some may remain weak and contribute only limited influence on the participating structures. Others may develop increasingly shared resonance organization, causing their behavior to become progressively interdependent. The distinction between weak, strong, and highly persistent relationships is therefore not regarded as fundamental, but as a consequence of the degree to which resonance organization becomes shared.

As relationships persist, they may themselves participate in further interactions and form new relationships with other structures. In this manner, interaction becomes a mechanism through which increasingly complex organizations can emerge. Relationship formation therefore provides the foundation for higher-order resonance structures and the hierarchical organization discussed in the following section.

2.4 Emergence and Hierarchical Organization

Every stable organization becomes a candidate for higher-order organization.

The formation of a relationship does not necessarily result in a new persistent identity. Many relationships may remain transient, while others persist only for limited durations. Within RQT, higher-order structures emerge when relationships acquire sufficient stability to participate in the resonance organization of their surroundings as identifiable entities in their own right.

The persistence of a relationship depends on the degree of compatibility between the participating structures. Compatibility is not regarded as a binary property, but as a condition that may develop through continued interaction. As structures influence one another, adjustments in their relative states may increase or decrease the stability of the relationship. Persistent relationships therefore arise not merely from interaction itself, but from the ability of interacting structures to accommodate one another sufficiently to maintain a shared organization over time.

When a relationship becomes persistent, a new level of identity emerges. This higher-order identity is not simply the sum of its constituents. Instead, it may exhibit resonance capabilities, interaction possibilities, and organizational properties that were not available to the participating structures independently. The emergence of such capabilities constitutes the basis of higher-order structure formation within the framework.

The formation of a new identity may also alter the accessibility of previously available relationships. Certain interactions may become enhanced, while others may become reduced, concealed, or effectively internalized within the newly formed structure. Emergence therefore introduces not only new possibilities for interaction, but also new forms of organization and selective accessibility.

Once established, higher-order identities may themselves participate in further relationship formation. The same principles that govern the emergence of lower-level structures may therefore operate recursively across multiple levels of organization. In this manner, increasingly complex resonance structures can arise without requiring additional foundational assumptions.

RQT consequently interprets hierarchy not as a separate organizing principle, but as a natural consequence of persistent relationship formation, emergent identity, and the continual development of new resonance capabilities across successive levels of organization.

2.5 Accessible and Enclosed Structure

Stability favors enclosure. Enclosure creates selective accessibility.

The emergence of higher-order structures introduces an important distinction between the internal organization of a structure and the properties it exposes to its surroundings. Within RQT, stable structures are not assumed to make all aspects of their internal resonance organization equally accessible. Instead, the formation of persistent identities naturally leads to varying degrees of accessibility between internal and external levels of organization.

This tendency arises from the continuous search for stable resonance configurations. As relationships become organized into higher-order structures, the participating components may reduce their dependence on external interactions by establishing increasingly stable internal relationships. The resulting organization can allow parts of the internal structure to become less directly involved in the structure's interactions with its surroundings.

Enclosure is therefore not regarded as a separate organizing principle, but as a natural consequence of stability. A structure that reduces unnecessary external dependencies may become less susceptible to disturbance and thereby maintain its identity more effectively over time. The degree of enclosure achieved by a structure depends on both its internal organization and the conditions imposed by its environment.

The emergence of enclosure creates selective accessibility. Some properties of a structure remain available for further relationship formation and interaction, while others become partially or fully internalized within the structure itself. As a result, the externally accessible behavior of a structure need not reveal the full complexity of its internal organization.

The accessible properties of a structure define how it participates in higher-order organization. Internal relationships, however, may continue to influence the behavior and stability of the structure even when they are not directly accessible from the outside. In this sense, every emergent structure may possess both an externally visible identity and an internal organization that extends beyond what is immediately available to its surroundings.

RQT therefore regards accessibility and enclosure as complementary aspects of stable organization. The formation of higher-order structures not only creates new identities and interaction possibilities, but also establishes boundaries between what remains externally available and what becomes incorporated into the internal organization of the structure.

2.6 Framework Assumptions and Scope

Complexity is not introduced. Complexity is accumulated.

The preceding sections have established the conceptual foundations upon which the Resonance Quantum Theory is developed. Rather than beginning with particles, forces, fields, or predefined physical laws, the framework starts from a small set of organizational assumptions and explores the consequences that follow from them.

RQT assumes the existence of a continuum capable of supporting oscillatory behavior. Within this continuum, persistent oscillatory configurations may arise and maintain their identity through self-resonance. Persistent structures may influence one another, form relationships, and develop increasingly stable forms of organization. Relationships that become sufficiently persistent may themselves emerge as higher-order identities capable of participating in further relationship formation.

The framework does not assume that all configurations persist. Persistence is itself regarded as a non-trivial property. Only structures that remain sufficiently stable can continue to participate in subsequent organization and relationship formation. Increasing levels of complexity may therefore arise through the continued accumulation of persistent relationships across multiple levels of organization.

From these assumptions emerge the concepts of identity, interaction, hierarchy, accessibility, and enclosure. Higher-order structures are understood not as independent organizing princi-

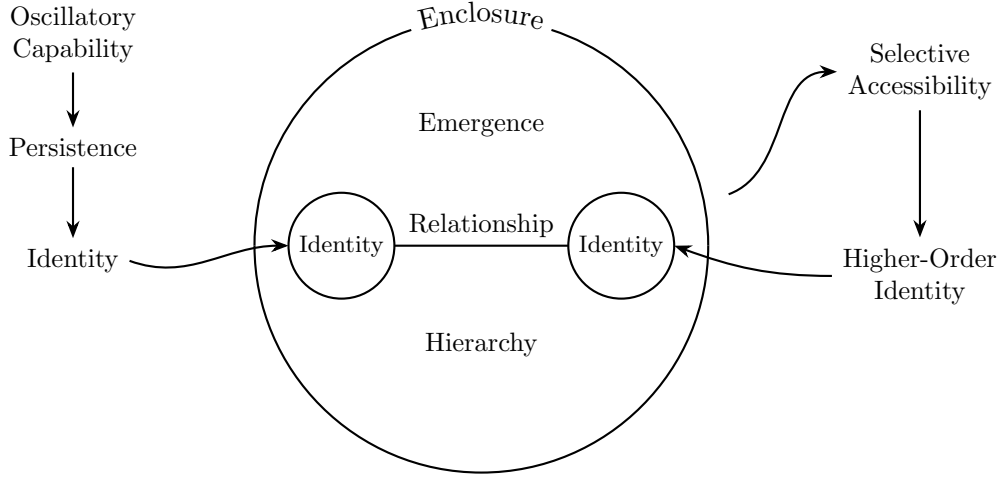


Figure 2: Conceptual foundations of the Resonance Quantum Theory

ples, but as consequences of recursive relationship formation and the development of persistent resonance organization.

At this stage, the framework intentionally refrains from specifying the detailed mechanisms through which such structures are realized. It does not initially assume particles, forces, charge, mass, spacetime geometry, probabilistic state descriptions, or specific interaction laws. These concepts, if required, are expected to emerge as effective descriptions of deeper resonance processes rather than as fundamental starting points.

The purpose of the framework is therefore not to introduce additional elementary entities, but to investigate how increasingly complex physical organization may arise from a minimal set of resonance-based assumptions. The following sections explore how specific resonance structures, interaction mechanisms, and observable physical phenomena may emerge from these principles.

3 The Resonance Quantum Theory

Conceptual foundations of the framework.

The conceptual foundations introduced the recursive emergence of persistent identities without assuming a specific physical realization. The Resonance Quantum Theory extends this framework by describing how such identities interact, organize, and evolve through resonance itself. The concepts introduced in this section remain independent of any particular geometric or discrete representation and therefore form the theoretical basis upon which the Roton Quantum Model is later constructed.

3.1 Resonance as a Fundamental Principle

Resonance is not treated in RQT as a special physical phenomenon occurring only under particular conditions. Instead, it is adopted as the fundamental principle by which oscillatory structures establish, maintain, and evolve stable organization.

The conceptual foundations introduced the emergence of persistent identities through recursive relationship formation without assuming a particular physical realization. The next step is to identify the principle that determines whether such relationships remain transient or develop into stable structures. RQT proposes that this principle is resonance.

In its most general sense, resonance describes the mutual adaptation of oscillatory systems whenever their evolving states become sufficiently compatible. Such adaptation may involve phase, characteristic frequencies, spatial organization, or other structural properties, without

requiring any specific geometric representation. While many interactions remain temporary, resonance provides the possibility for oscillatory systems to organize into persistent configurations.

From the perspective of RQT, resonance is therefore not regarded as a consequence of existing physical objects. Rather, stable physical objects are understood as consequences of persistent resonance. Resonance is treated as the organizing principle from which increasingly stable structures may emerge, interact, and recursively form higher-order identities.

The following sections develop this framework by introducing the continuum in which resonance occurs, the formation of localized resonance structures, and the mechanisms by which resonance accommodates, stabilizes, and eventually gives rise to observable physical systems.

3.2 The Resonance Continuum

RQT does not begin by asking what space is made of. It begins by asking what is minimally required for stable resonance to exist.

RQT does not attempt to explain the origin or fundamental nature of the substrate in which resonance occurs. Instead, it adopts the minimal assumption that a continuum exists which is capable of supporting persistent oscillatory organization. Beyond this requirement, no particular physical interpretation is assumed.

From the perspective of RQT, this continuum should not initially be identified with physical space or spacetime. Rather, it represents the possibility for oscillatory structures to emerge, persist, and mutually influence one another. Observable notions such as separation, spatial distance, and temporal evolution are not taken as fundamental prerequisites but are considered possible emergent descriptions of stable resonance organization.

This represents a conceptual inversion of the usual viewpoint. Rather than assuming that resonance occurs within a pre-existing spacetime, RQT explores the possibility that the observable notions of space and time arise from the formation, separation, and continued interaction of persistent resonance identities. The theory therefore begins with distinction and relationship rather than with metric geometry.

Only one assumption is required: sufficiently compatible oscillatory structures may establish persistent relationships. These relationships provide the basis from which increasingly organized resonance structures may emerge and subsequently interact through newly available resonance pathways. The concepts of distance, propagation, and geometry are introduced only insofar as they become meaningful descriptions of these emerging relationships.

A consequence of this viewpoint is that the underlying mechanism giving rise to the first persistent oscillatory capability need not itself remain directly accessible. If higher-order resonance structures arise through successive stages of resonance organization, then the deeper levels from which these structures originate may become intrinsically concealed by the same principles of resonance confinement that later govern stable physical systems. In this sense, the apparent simplicity of a persistent oscillation may itself be the observable residue of a deeper resonance organization whose internal structure no longer participates directly in accessible interactions.

The following sections therefore take the existence of this resonance continuum as a conceptual starting point and investigate how localized resonance structures emerge, interact, accommodate one another, and eventually give rise to stable physical organization.

3.3 Distinct Resonance Structures

The first observable property of a resonance structure is not its location, but its persistent distinction.

The existence of a resonance continuum alone does not imply the existence of distinguishable structures. Many oscillatory configurations may remain transient, continuously emerging and dissolving without establishing persistent organization. RQT therefore distinguishes between

arbitrary oscillation and oscillatory organization that becomes sufficiently persistent to form a distinct resonance structure.

A distinct resonance structure is not initially defined by position, geometry, dimensionality, or physical extent. Instead, it is defined by the continued persistence of its oscillatory organization and by its distinguishability from the surrounding resonance continuum. Distinction therefore precedes localization. Only after stable relationships emerge may geometric descriptions become meaningful.

Persistent distinction also implies continuity. A resonance structure possesses both a history and the potential for continued evolution, not because information is explicitly stored, but because its oscillatory organization remains continuously connected to its own past. The present state is therefore understood as the ongoing evolution of the same resonance process rather than as a sequence of independently recreated states.

From the perspective of RQT, a resonance structure remains part of the observable resonance organization only insofar as this continuity persists. A structure that became completely disconnected from all continuing resonance relationships would cease to participate in the same observable organization. Whether such complete decoupling is possible remains an open question. Everyday physical structures, however, exhibit persistent relationships over extended periods, suggesting that stable resonance organization generally preserves continuity rather than abandoning it.

RQT begins not with localized objects, but with persistent distinction. Identity emerges from the continued existence of a persistent resonance organization. Interaction, separation, and geometry emerge only thereafter through the relationships established between such structures.

Distinct resonance structures therefore provide the first entities capable of participating in persistent interaction. The following section examines how these structures influence one another, adapt through resonance, and establish the relationships from which increasingly organized physical systems emerge.

3.4 Resonance Interaction

Interaction begins whenever the continued evolution of one distinct resonance structure influences that of another.

The existence of distinct resonance structures inevitably allows the possibility of interaction. Whenever two structures participate in one another's continued evolution, they no longer develop independently. Their oscillatory organization begins to adjust through mutual influence, giving rise to resonance interaction.

Interaction should not be understood as a discrete event but as a continuous process. Some interactions remain transient, producing only temporary adjustments before the participating structures continue independently. Others persist, allowing the interacting structures to accommodate one another over extended periods. Resonance interaction therefore represents a dynamic process whose consequences depend on the continued compatibility of the participating structures rather than on the mere occurrence of a single encounter.

Accommodation does not necessarily imply observable change. Highly compatible resonance structures may remain in continuous interaction while exhibiting little or no apparent evolution because their oscillatory organization is already mutually consistent. Interaction therefore exists whenever resonance structures contribute to one another's continued organization, regardless of whether this contribution manifests as rapid adaptation or as stable equilibrium.

The degree to which an interaction contributes to the continued organization of the participating structures determines its significance within the evolving resonance system. Persistent interactions gradually become part of the resonance organization itself, while transient interactions leave little lasting influence. The following sections investigate how such persistent interactions organize into stable resonance relationships and eventually give rise to the structured resonance pathways that characterize stable physical systems.

3.5 Resonance Accommodation

Interaction becomes organization when resonance structures begin to accommodate one another.

Resonance interaction is not a fixed relation between otherwise unchanged structures. Continued interaction influences the evolution of all participating resonance structures and may gradually alter the compatibility of their oscillatory organization. RQT refers to this process as resonance accommodation.

Accommodation is understood as a continuous and mutual process. The participating structures do not merely act upon one another from the outside; their evolving resonance organization becomes increasingly interdependent. This may involve adjustments in phase, characteristic periods, relative organization, or other resonance properties, without requiring any particular geometric representation at this stage.

Successful accommodation allows interacting structures to develop a partially shared resonance evolution. Their future states can no longer be described independently, because each structure contributes to the continued organization of the other. Where compatibility increases, the interaction may persist and become part of a larger resonance organization. Where compatibility cannot be maintained, the interaction may remain transient, weaken, or lead to reorganization.

Accommodation does not necessarily produce visible change. Two highly compatible structures may remain in continuous interaction while exhibiting little apparent evolution, precisely because their resonance organization is already mutually consistent. Stable resonance therefore represents not the absence of interaction, but a condition in which further accommodation is only minimally required.

In realistic systems, accommodation is not limited to isolated pairs. Every resonance structure may participate in several simultaneous interactions, each contributing to its continued evolution. Resonance accommodation therefore occurs within a network of mutually constraining relationships, where the persistence of any one interaction depends on its compatibility with the wider resonance organization.

RQT does not require that resonance structures consciously or globally optimize their organization. It proposes only that interactions capable of maintaining compatible shared evolution remain available for continued organization, while incompatible interactions tend to remain transient or reorganize. In this sense, accommodation does not create resonance; it reveals which resonance relationships can persist.

The persistence of accommodated interaction provides the conceptual basis for the structured resonance relationships introduced in the following sections.

3.6 Stability and Persistence

Stability denotes the persistence of an organized resonance process, not the absence of change.

A resonance organization becomes stable when its continued evolution preserves the distinction and internal coherence by which it can be identified. Such stability does not require every oscillatory state or resonance relationship to remain unchanged. Rather, the organization must retain sufficient continuity for its present evolution to remain connected to its preceding states.

Even a simple resonance relationship contains both internal and external aspects. Internally, the participating structures sustain the mutual organization through which they form a distinguishable system. Externally, each participating structure may remain exposed to additional resonance influences arising from its surrounding environment or from relationships at another level of organization. Stability therefore depends not only on internal compatibility, but also on the capacity of the system to accommodate changing external demands.

External or internal changes may introduce resonance mismatch into an existing organization. A stable system can respond by adjusting and redistributing this mismatch across its participating relationships while preserving the organization as a whole. The resulting changes may occur

within subordinate structures, within the relationships that hold them together, or in the system's external relations, without necessarily interrupting its persistent identity.

Stability is therefore dynamic rather than static. A stable resonance organization may undergo continuous internal adjustment while remaining distinguishable as the same continuing process. The more effectively such adjustments remain bounded within the existing organization, the less they disturb its higher-level identity.

Persistence and stability describe related but distinct aspects of this continuation. Persistence refers to the continued existence of a distinguishable resonance organization. Stability refers to its capacity to preserve that continued identity under changing resonance conditions.

When an imposed resonance mismatch can no longer be accommodated within the existing organization, the system may reorganize, separate, or dissolve into configurations capable of more persistent resonance evolution. Separation is therefore not necessarily the failure of accommodation, but may itself restore compatibility among the remaining resonance structures.

A stable resonance system is thus not isolated from change. Its stability lies in maintaining a recognizable organization while continuously accommodating the resonance influences in which it participates.

3.7 Resonance Distribution Across Hierarchies

Resonance organization is sustained not only by local accommodation, but by the distribution of accommodation demands across nested structures and relationships.

A distinguishable resonance structure may itself contain subordinate resonance structures, while simultaneously participating in larger resonance organizations. RQT therefore considers physical organization to be hierarchical: an identity may act as a coherent whole at one level while remaining internally composed of interacting sub-identities at another.

Changes introduced at one level need not remain confined to that level. An external resonance influence may alter a relationship within the larger identity, while the resulting mismatch is redistributed through its internal organization. Conversely, changes arising within a subordinate structure may propagate outward and affect the relationships through which the higher-level identity is maintained.

The response of a resonance system therefore depends on the distribution of its available accommodation capacity. Some disturbances may be absorbed locally, requiring only minor adjustment within a limited part of the structure. Others may be distributed across several internal relationships, producing a broader but still coherent reorganization. Where the existing organization cannot accommodate the imposed mismatch, the disturbance may extend to the system boundary and alter its external relationships.

This distribution is not assumed to follow a predetermined geometric pathway. It reflects the evolving compatibility of the resonance relationships participating in the system. Accommodation proceeds through those parts of the organization capable of contributing to a more persistent shared evolution, while strongly incompatible relationships may weaken, reorganize, or separate.

A higher-level identity may therefore remain stable even while its internal resonance organization changes substantially. Its persistence depends not on the immutability of its components, but on whether the combined adjustments preserve the relationships through which the whole remains distinguishable. In this sense, stability at one level may be supported by continuous reorganization at another.

Resonance structures are also constrained by their wider environment. Because each identity participates in several simultaneous relationships, no accommodation occurs in isolation. A configuration compatible with one relationship may introduce mismatch into another. Persistent organization therefore emerges from the distributed accommodation of multiple, mutually constraining resonance influences.

Where accommodation succeeds, previously distinct structures may participate in a more coherent higher-level identity. Where it fails, the existing organization may release subordinate

structures, alter its external relationships, or separate into new resonance systems. Formation and dissolution are thus complementary outcomes of the same underlying process: the redistribution of resonance toward configurations capable of continued organization.

This hierarchical view prepares the transition from general resonance processes to the structured relationships through which persistent resonance organization can be represented. The following sections introduce the resonance channels, closures, and enclosures that describe these enduring forms of shared resonance evolution.

4 Resonance Channels and Closure

The formation of observable structure.

The preceding sections described resonance organization as a continuing process of interaction, accommodation, redistribution, and persistence. As the number of participating structures and relationships increases, however, a purely process-based description becomes increasingly difficult to follow. A compact representation is therefore required for the persistent organization produced by these processes.

RQT introduces the concepts of resonance channels and resonance closure for this purpose. These concepts do not add new physical ingredients to the theory. A resonance channel represents the continuing organization of a persistent interaction, while closure describes the extent to which that interaction has become incorporated into a self-consistent resonance structure.

A channel is therefore not assumed to be a material conduit or a geometrically defined path. At the general level of RQT, it expresses a persistent resonance relationship whose organization contributes to the identities of the participating structures and, where applicable, to the higher-level identity they form together.

Not every resonance relationship becomes fully closed. Some remain partially accommodated, externally exposed, or distributed across wider resonance systems. The distinction between open, residual, and closed resonance organization will later allow increasingly complex systems to be described without requiring their complete internal evolution to be followed at every level.

These concepts remain independent of any particular microscopic geometry. The Roton Quantum Model will later propose specific structural representations of resonance channels, nodes, closures, and enclosures. At this stage, channels and closure are introduced only as the general relational language through which persistent resonance organization can be described.

4.1 Resonance Channels

A resonance channel is the persistent organization of a distinguishable resonance interaction.

Resonance interaction may be transient, diffuse, distributed, or continuously changing. Where such interaction becomes sufficiently persistent and organized to contribute to the continuing identity of a resonance structure or system, RQT represents it as a resonance channel.

A channel is therefore not an additional physical substance, conduit, or pre-existing path through space. It is the relational organization through which resonance structures participate in a shared evolution. The interaction is fundamental; the channel is its persistent and distinguishable form.

A resonance channel need not initially possess a defined geometric length, width, curvature, or orientation. At the general level of RQT, it is characterized by the persistence, compatibility, and organization of the interaction it represents. Specific geometric properties may arise only when the channel is embedded within a more detailed structural model.

Channels may differ in the way they participate in a resonance system. Some form persistent relationships between identifiable counterparts. Others remain externally exposed, distributed across several interactions, or only partially organized around a dominant partner. A channel

may therefore be open, relationally occupied, redirected, or incorporated into a more completely closed resonance organization.

The existence of a channel also does not require an unchanging interaction partner. What persists may be the organized mode of coupling itself, while the structures contributing to it change over time. This allows a channel to represent both stable pairwise relationships and persistent interactions with a distributed resonance environment.

A realized channel contributes to the identities of the structures it connects. Where several resonance structures become organized through persistent channels, their relationships may together establish a distinguishable higher-level resonance identity. Channels therefore do not merely connect pre-existing objects; they participate in defining what those objects and larger systems become.

The number and mutual compatibility of channels available to a given identity are not determined by general RQT alone. They depend on the adaptability and internal organization of the resonance structure supporting them. The Roton Quantum Model later represents these constraints through specific channel types, channel counts, and structural arrangements.

4.2 Open and Closed Channels

A realized resonance channel may remain open or become increasingly closed according to how fully its interaction is incorporated into the resonance organization under consideration.

An open resonance channel forms a persistent relationship between distinguishable resonance structures, but remains at least partly available to further accommodation. Its organization may still respond substantially to external resonance influences, shift between competing relationships, or participate in the formation of a larger resonance system. Openness therefore does not imply that the channel is weak or temporary. It indicates that the relationship has not yet become fully internalized within a stable resonance organization.

A closed resonance channel is a persistent interaction that has become predominantly incorporated into the internal organization of a resonance system. Closure does not terminate or deactivate the relationship. Rather, the participating structures sustain one another through an increasingly self-consistent resonance process, making the channel part of the identity of the resulting system.

The distinction between open and closed channels is not necessarily binary. Continued accommodation may gradually increase closure, while changing resonance conditions may partially or temporarily reopen a previously closed channel. During such reopening, the system becomes more accessible to external accommodation and may redistribute, redirect, or reorganize the relationship.

Whether a channel is considered open or closed also depends on the identity level being examined. A channel may be closed within one resonance system while the resulting higher-level identity remains open to additional relationships of its own. Closure at one level therefore does not imply isolation from the wider resonance environment.

By becoming closed, a channel changes more than its own relational state. Its incorporation alters the resonance organization of the participating structures and thereby changes which further relationships remain compatible and available to the resulting identity.

Closure changes the future compatibility landscape of the resulting identity.

4.3 Residual Resonance

Residual resonance is resonance influence that remains externally accessible without being fully organized into a closed relationship.

Residual resonance may arise where a realized channel is only incompletely closed, temporarily reopened, or continuously affected by redistribution within its enclosing structure. In such cases,

part of the interaction remains accessible to the wider resonance environment even though the principal relationship is sustained internally.

Residual resonance may also arise from an available channel mode that has not become organized around a distinct counterpart. Its influence then remains distributed across the surrounding resonance environment rather than forming a directed, partner-specific channel. Another compatible structure may respond to this distributed influence and, through continued accommodation, establish a realized resonance channel.

Residual resonance therefore differs from a closed channel both in organization and relational specificity. A closed channel sustains a distinguishable relationship within an enclosure, whereas residual resonance may remain diffuse, externally exposed, and shared among several possible interactions.

The strength and distribution of residual resonance may later be related to the emerging separation and geometry of resonance structures. Their quantitative representation, including possible distance-dependent laws, belongs to the more specific development of the Roton Quantum Model.

4.4 Resonance Closure

Resonance closure occurs when a persistent interaction becomes sufficiently accommodated to form an internally sustaining part of a resonance identity.

A realized resonance channel may remain externally exposed, continue to compete with alternative relationships, or require ongoing adjustment between its participating structures. Through continued accommodation, however, the interaction may become increasingly self-consistent. RQT describes this progressive internalization of a resonance relationship as resonance closure.

Closure does not terminate the interaction. On the contrary, the interaction remains active as part of the continuing organization through which the participating structures sustain one another. What changes is its relational role: an externally negotiable interaction becomes an internally supporting component of a larger resonance system.

Resonance closure need not occur instantaneously or absolutely. A relationship may become partially closed while retaining some susceptibility to external disturbance, redistribution, or partner change. Changing resonance conditions may temporarily reopen or reorganize a previously closed channel, allowing new accommodation before the relationship either returns to closure or develops into a different configuration.

Closure also changes the resonance possibilities of the participating structures. As a channel becomes incorporated into a stable organization, the modes previously available for equivalent external coupling may be reduced, redirected, or removed. Other nearby structures may therefore fail to establish a persistent channel, not because interaction has ceased altogether, but because the closed organization no longer presents a sufficiently compatible resonance mode.

The effect of closure depends on the identity under consideration. Closing one channel may saturate all available relationships of a particular type, or it may leave other compatible channels accessible. The specific number and arrangement of such possibilities belong to the internal structure of the identity and are developed more concretely within the Roton Quantum Model.

Closure should therefore be understood neither as perfect isolation nor as the disappearance of residual influence. Even a strongly closed relationship may remain weakly affected by its wider resonance environment, and small redistributions may temporarily increase its external accessibility. Closure instead denotes the condition in which the relationship is predominantly sustained within the internal organization of the resulting resonance system.

By converting a persistent interaction into an internally sustaining relation, resonance closure prepares the emergence of a higher-level identity. The resonance enclosure formed through such closed relationships is considered in the following subsection.

4.5 Channel Availability and Saturation

4.6 Resonance Enclosure

4.7 Resonance Locking

5 Information, Stability and Inertia

5.1 Information as Resonance Structure

5.2 Structural Persistence

5.3 Accommodation Dynamics

5.4 Rotonal Inertia

5.5 Hierarchical Stability

6 Hierarchical Resonance Structures

6.1 Resonance Nodes

6.2 Resonance Networks

6.3 Resonance Clusters

6.4 Emergent Hierarchies

7 The Roton Quantum Model

Discrete representation of resonance structures.

7.1	Purpose of the Model
7.2	Rotons
7.3	Sonons
7.4	Resonons
7.5	Resonance Centers
7.6	Linear Resonance Channels
7.7	Model Limitations
8	Elementary Resonance Structures
8.1	Photons
8.2	Electrons
8.3	Positrons
8.4	Electron-Positron Systems
8.5	Neutrino-like Structures
9	Nuclear Resonance Structures
9.1	Protons
9.2	Neutrons
9.3	Alpha Structures
9.4	Nuclear Resonance Networks
9.5	Binding Energy and Closure
10	Atomic and Molecular Structure
10.1	Atoms
10.2	Molecular Binding
10.3	Resonance Networks
10.4	Closure Hierarchies
11	Interactions and Force Emergence
11.1	Residual Resonance Effects
11.2	Electromagnetic Interaction
11.3	Gravitational Interaction
11.4	Long-Range Resonance Coupling
12	Mathematical Representations
12.1	Continuous Descriptions
12.2	Discrete Descriptions
12.3	Field Approximations
12.4	Numerical Simulation Approaches
13	Predictions and Open Questions

15 Conclusions

Summary and outlook.

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Companion Website and Further Material

A non-technical introduction, extended explanations, graphical illustrations and ongoing discussions of the Roton Quantum Model are available on the project website:

- Roton Quantum Model – intuitive overview and guided tour: <https://ledoigt.ch>
- Technical notes, simulations and updates: <https://ledoigt.ch/ledoigt/notes>
- Discussion and feedback channel: <https://ledoigt.ch/ledoigt/discuss>

These resources are intended as a more accessible companion to the present article, offering step-by-step motivation, visualizations, and evolving refinements of the model.

Data and Code Availability

Simulation scripts, illustrative numerical experiments and example configurations of Rotons used to explore the qualitative behavior reported in this work are (or will be) made available at:

- Code repository (numerical experiments): <https://github.com/ledoigt>
- Supplementary animations and figures: <https://ledoigt.ch/ledoigt/collections/>

Interested readers are invited to reproduce, extend and critically test the presented ideas.

Author’s Note

This work is intended as an open invitation to the theoretical physics and complex-systems communities to examine, challenge and refine the Roton Quantum Model. The author explicitly welcomes critical assessments, alternative formulations and concrete proposals for experimental or numerical tests, which may help to clarify the model’s scope, limits and possible connections to established frameworks.